

120 Days of Quantitative Finance

Month 4: Stochastic Calculus

Week 14: SDEs and Black-Scholes

Day 96: Option Pricing under Black-Scholes and
Jump-Diffusion

1 Introduction

The Black-Scholes model assumes that asset prices evolve continuously. However, real markets often exhibit sudden jumps.

To address this, we extend the model to include jump processes.

2 Black-Scholes Model

Under Black-Scholes, the stock price follows:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

The option price is obtained using a closed-form solution.

3 Limitations

Black-Scholes assumes:

- continuous paths

- constant volatility
- no sudden price jumps

These assumptions often fail in practice.

4 Jump-Diffusion Model

To incorporate jumps, we use the Merton Jump-Diffusion model:

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_t(J - 1)dN_t$$

where:

- N_t is a Poisson process (jump arrival)
- J is the jump size

5 Interpretation

- diffusion term models normal fluctuations
- jump term captures sudden market shocks

6 Monte Carlo Pricing

Option prices can be estimated by:

1. Simulating multiple price paths
2. Computing payoff at maturity
3. Discounting expected payoff

7 Comparison

- Black-Scholes produces smooth price paths
- Jump-diffusion produces discontinuities
- Jump models better capture real market behavior

8 Conclusion

While Black-Scholes provides analytical simplicity, jump-diffusion models offer a more realistic representation of financial markets.